Pigeonhole Principle

The pigeonhole principle states that if k+1 or more pigeons are placed into k pigeonholes then there is at least one pigeonhole containing two or more of the pigeons.

Proof: We use proof by contradiction here. Suppose that k+1 or more boxes are placed int k boxes and no boxes contain more than one object in it. If there are k boxes then there must be k objects such that there are no two objects in a box. This contradicts that our assumption. So there is at least one box containing two or more of the objects.

Example 1: Among any group of any 367 people, there must be at least two with the same birthday because there are only 366 possible birthdays.

Example 2: In any group of 27 English words, there must be at least two that begin with the same letter, because there are 26 letters in the English alphabet.

Theorem: The Generalized Pigeonhole Principle

If N objects are placed into k boxes, then there is at least one box containing at least |N/k| objects.

Proof: We will use a proof by contradiction. Suppose that none of the boxes contain more than |N/k|-1 objects. Then, the total number of objects is at most

$K(\lceil N/k \rceil-1)<k((N/k+1)-1) = N.$

Where the inequality $\lceil N/k \rceil<(N/k)+1$ has been used. This is contradiction because there total of N objects.

A common type of problem asks for the minimum number of objects such that at least r objects of these objects must be in one of k boxes when these objects are distributed among the boxes. When we have N objects, the generalized pigeonhole principle tells us there must be at least r objects in one of the boxes as long as $\lceil N/k \rceil \geq r$. The smallest integer N with N/k>r-1, namely N = k(r-1)+1, is the smallest integer satisfying the inequality $\lceil N/k \rceil \geq r$. Could a smaller value of N suffice? The answer is no, because if we have k(r-1) objects, we could put r-1 of the them in each of the k boxes and no box would have at least r objects.

Example 1: Among 100 people there are at least $\lceil 100/12 \rceil=9$ who were born in the same month.

Example 2: What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades A, B, C, D and F?

Solution: The minimum number of students needed to ensure that at least six students receive the same grade is the smallest integer N such that $\lceil N/2 \rceil = 6$. The smallest such integer N = 5.5+1 = 26. If you have only 25 students, it is possible for there to be five who have received each grade so that no six students have received the same grade. This, 26 is the minimum number of students needed to ensure that at least six students will receive the same grade.

Permutations

Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of particular size, where the order of these elements matters. Many other

counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the order of the elements selected does not matter. For example, in how many ways can we select three students from a group of five students to stand in line for a picture? How many different committees of three students can be formed from a group of four students?

Permutation: A permutation of a set of distinct objects is an ordered arrangement of these objects. We also are interested in ordered arrangements of some of the elements of a set. An ordered arrangement of r elements of a set is called an r-permutation.

Theorem

If n is a positive integer and r is an integer with 1≤r≤n then there are
p(n,r) = n(n-1)(n-2)……(n-r+1) r-permutations of a set with n distinct elements.

Corollary: If n and r are integers with 0≤r≤n then P(n,r) = $\frac{n!}{(n-r)!}$

Example 1: Show that (n-r+1) .$^{n}P_{r-1}$ = $^{n}P_{r}$

Solution:

LHS = (n-r+1)$^{n}P_{r-1}$

= (n-r+1)*$\frac{n!}{[n-(r-1)]!}$

=(n-r+1)*$\frac{n!}{[n-r+1]!}$

= (n-r+1)*$\frac{n!}{(n-r+1)*(n-r)!}$

= n!/(n-r)!

=$^{n}P_{r}$

Example 2: If 56Pn+6 :54Pn+3 = 30800:1 then find n

Solution:

We have $^{n}P_{r}$ = $\frac{n!}{(n-r)!}$

So $^{56}P_{n+6}$ = $\frac{56!}{(56-n)!}$

Similarly,

$^{54}P_{n+3}$ = $\frac{54!}{(50-n)!}$

Now, given,

$^{56}P_{n+6}$: $^{54}P_{n+3}$ = $\frac{56!}{(56-n)!}$: $\frac{54!}{(50-n)!}$ = 30800:1

n = 41

Example 3: How many license plates consisting of 3 digits can be made out of given integers 3,4,5,6,7?

Solution: This is just like arranging 3 objects out of 5 objects, So we have,

P(5,3) = 5!/(5-3)! = 60

The problem can also be solved by product rule.

Example 4: How many number of 3 digit can be formed from the digits 3, 4,5, 6, 7, 8? How many of these are divisible by 5.

Solution: Here, n=6, r=3

Therefore P(6,3) = 6!/(6-3)! = 120

To find the number divisible by 5, we fix digit 5 on unit place

Therefore n = 6-1 = 5 and r = 3-1 = 2

Therefore P(5,2) = 5!/(5-2)! = 20

Permutations with Repetitions

The permutations of **n** objects taken all at a time when there are **p** objects of one kind, **q** objects are of second kind, **r** objects are of third kind, is $\frac{n!}{p!q!r!}$

Example 8 How many sever-letter words can be formed using the letters of word "BENZENE"?

Solution: There are seven letters of which there are three Es and two Ns. Therefore, the number of permutations is $\frac{7!}{3!2!}$= 420

Example 9: How may six-letter words can be made using the letters of the word "SUNDAY"

Solution: There are six-letters of which none is repeated. Therefore, the number of permutations is $\frac{6!}{0!}$ =6.5.4.3.2.1=720

Circular Permutation

The number of ways of arranging n unlike objects in a ring when clockwise and anticlockwise arrangements are different is (n-1)!. For example, consider 4 people A,B,C and D who are to be seated at a round table. To find the number of different arrangements, we fix A and then consider the number of ways of arranging B, C and D. Therefore the number of different arrangements of 4 people around the table is 3!.

The number of ways of arrangement n unlike objects in a ring, when clockwise and anticlockwise arrangement are same and is (n-1)!/2.

Example 14 : In many ways can the numbers on a clock face be arranged?

Solution: In a clock face there are 12 numbers. So they can be arranged in (12-1)! = 11 ways.

Repeated Use of the Same Objects

The number of distinct arrangements of n objects taken r at a time with repetition is n^r

Example 16: In how many ways can a garland of 15 different flowers be made?

Solution: The arrangement of 15 different flowers to make a garland, in both clockwise and anticlockwise direction is same. So, they can arranged in (15-1)/2 = 14!/2 ways

Combination

Combination of objects means just their collection without any regard to order or arrangement. An unordered selection of r objects from a set of n objects is called a combination or r-combination. It is denoted by C(n,e) or ^nC_r and is given by the expression.

C(n,r) = $\frac{n!}{r!(n-r)!}$

Example 18: Calculate the value of C(12,10)

Solution: C(12,10) = $\frac{12!}{10(12-10)!}$ = 66 ways

Example 19: In how many ways can a hand of 4 cards be dealt from an ordinary pack of 52 playing cards?

Solution: We need to consider combinations, since the order in which the cards are dealt is not important.

Now, C(52,4) = $\frac{52!}{4!(52-4)!}$ = 270725 ways.

Permutations with Indistinguishable Objects

The number of different permutations of n objects, where there are n1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2…….and n_k indistinguishable objects of type k is

$$\frac{n!}{n1!\,n2!\,n3!\,\ldots\ldots n3!}$$

Example 24: How many different strings can be made by recording the letters of the word **SUCCESS?**

Solution: 7! /3!2!1!1! = 5040 /12 = 420 (by a theorem in the book) because there are three Ss, two Cs, one U, and one E. Another way: there are (7 3)ways to choose 3 positions for Ss, then (4 2) ways to choose 2 positions for Cs, then 2 ways to choose the position for U, and the remaining letter must be E. There are (7 3).(4 2).2 = 7! /3!4! .4! /2!2!.2 = 35 · 6 · 2 = 420.

Binomial Theorem

Let x and y be two variables and n be non-negative integers then the binomial theorem for positive index n states that

$(x+y)^n = C_0(n,0)x^n+C_1(n,1)x^{n-1}y^1+C_2(n,2)x^{n-2}y^2+\ldots\ldots\ldots\ldots\ldots\ldots+C_n(n,n)y^n$

The coefficient C_0, C_1, $C_2\ldots C_n$ are called binomial coefficients.

General Term

The $(r+1)^{th}$ term in the expansion of $(x+y)^n$ is usually called its general term which is denoted by t_{r+1}

In the expansion of $(x+y)^n$

$t_1 = 1^{st}$ term $= C(n,0)x^n$

$t_2 = 2^{nd}$ term $= C(n,1)x^{n-1}y^1$

$t_3 = 3^{rd}$ term $= C(n,2)x^{n-2}y^2$

$t3 = 4^{th}$ tem $= C(n,3)x^{n-3}y^3$

….

$t_{r+1} = (r+1)^{th}$ term $= C(n,r)x^{n-r}y^r$.

Thus, the general term in the expansion of $(x+y)^n$ is t^{r+1} which is $C(n,r)x^{n-r}y^r$

Example 25: Expand $(2a+5)^5$ by the binomial theorem

Solution: here n=5

By using binomial theorem,

$(2a+5)^5 = C(5,0)(2a)^5+C(5,1)(2a)^{5-1}(5)^1+C(5,2)(2a)^{5-2}(5)^2+C(5,3)(2a)^{5-3}(5)^3+C(5,4)(2a)^{5-4}(5)^4+C(5,5)(2a)^{5-5}(5)^5$

$= 32a^5+400a^4+2000a^3+5000a^2+6250a+6250$

Example 26: Find the coefficient of the term containing y8 in the expansion of $(x+3y^2)^{17}$

Solution: The term containing y8 in the expansion of $(x+3y^2)^{17}$ is $C(17,4)x^{17-4}(3y^2)^4 = 2380x^{13}.81y^8$

The coefficient of the term containing y8 in the expansion $(x+3y^2)^{17} = 192780$

Example 27: Find the coefficient of x^{16} in the expansion of $(2x^2-x/2)^{12}$

Solution:

Comparing $(2x^2-x/2)^{12}$ with $(x+y)^n$ we get, $x = 2x^2$, $y = x/2$ and n=12

The general term in the expansion of this expression by binomial theorem is

$t_{r+1} = C(12,r)(2x^2)^{12-r}(-x/2)^r$

$= C(12,r)2^{12-r}(-1/2)^r x^{2(12-r)+r}$

$= C(12,r)2^{12-2r}(-1)^r x^{24-r}$

Then the coefficient of x^{16}, we put 24-r = 16 thus r=8

Then the coefficient is $C(12, 8)2^{12-16}.(-1)^8$

$= C(12,8).(1/2)^4$

495/16

Middle Term

Now let us find the middle term or terms in the expansion of $(x+y)^n$. we have to consider the case when n is even number and when n is odd number.

When n is even number: When n is even, the number of terms in the expansion of $(x+y)^n$ is n+1 which is odd. So there is exactly one middle term which is $t_{n/2+1}$

When n is odd: When is n is odd, the number of terms in the expansion of $(x+y)^n$ is n+1 which is even. So there are two middle terms which are $t_{(n+1)/2}$ and $t_{(n+1)/2+1}$

Pascal's Triangle

The geometrical arrangement of binomial coefficients in the expansion of (x+y)n in a triangular form is called Pascal's triangle. This triangle array of numbers is called Pascal's triangle.

The coefficients of the successive powers of a+b can be arranged in a triangular array of numbers which can be shown as:

Coefficients in $(a+b)^0$				1			
Coefficients in $(a+b)^1$			1		1		
Coefficients in $(a+b)^2$		1		2		1	
Coefficients in $(a+b)^3$	1		3		3		1

The terms in Pascal's triangle have the following properties:

a. The first and last number in each row is 1
b. Every other number in the array can be obtained by adding the two numbers appearing directly above it.

Example 32: Write down the expansion of $(1+y)^6$ using Pascal's Theorem

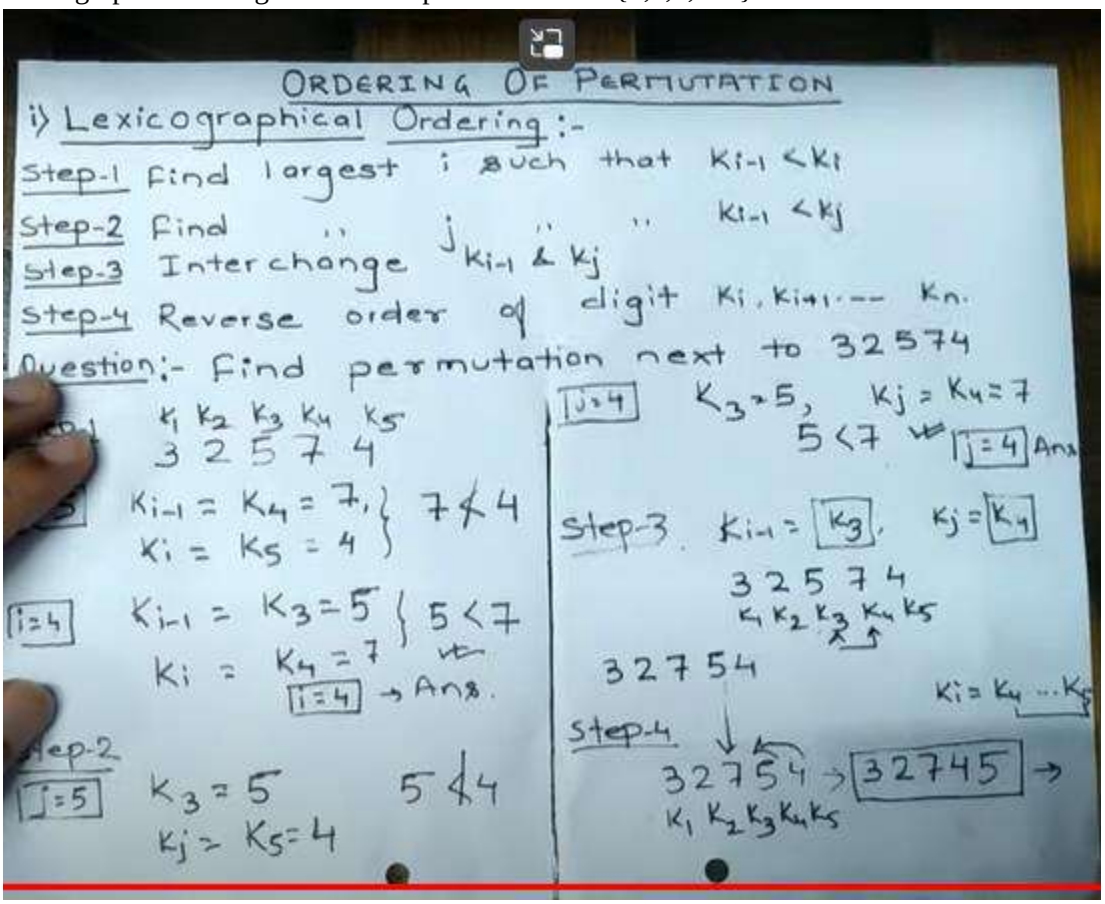
Solution: Here n = 6, so we use Pascal's triangle up to n=6

When n=0						1						
When n=1					1		1					
When n=2				1		2		1				
When n=3			1		3		3		1			
When n=4		1		4		6		4		1		
When n=5	1		5		10		10		5		1	
When n=6	1	6		15		20		15		6		1

Generating Permutations and Combinations

Any set of n-elements can be mapped one to one with the set {1,2,3,…n}. We can find the permutation of any set of n elements by generating the permutation of the n-smallest positive integers and then replacing these integers with the corresponding elements.

To generate permutations of a set of a elements there are many algorithms. One of these algorithm is lexicographic ordering of the set of permutations of {1,2,3,4..n}.



Algorithm

Given a natural number 'n' enlist n! Permutations of given n distinct objects say 1,2,3… in ascending order.

Steps

1. Let C = 1, print first permutations of 1,2,3…n. If n=1 then go to set 3
2. For C = 2 to "Total Count", Generate next permutation from permutation $a_1,a_2,a_3…a_n$ by following steps:
    a. Scan the digits of given permutation from right to left and note first consecutive pairs (a_{n-1} and a_n) such that $a_{n-1}<a_n$
    b. Search for the smallest digit among digits a_{m+1}, a_{m+2},….a_n that is longer than a_m call it x
    c. Interchange a_m and X
    d. Arrange all digits a_{m+1} to an in increasing order as $a_{m+1}<a_{m+2}<a_n$
    e. Print a_1,a_2, a_3…….a_n
3. Stop

Example 1: Generate all permutations of 1, 2, 3 and 4

Solution:

1234	1243	1324	1342	1423	1432	2134
2143	2314	2341	2413	2431	3124	3142

3214 3241 3412 3421 4123 4132 4213
4231 4312 4321

Example 2: Generate the permutation of the set {Physics, Chemistry, Math, biology}.

Solution: To generate permutation of set S {Physics, Chemistry, math, biology} we first can generate the permutation of the set [1, 2, 3, 4} and the maps

1. Physics
2. Chemistry
3. Maths
4. Biology

Generating Combination

Algorithm

[Select k objects of n distinct, say 1, 2, 3, 4…..n]

1. List all a-digit in ascending order
2. Select first k-digit , print them
3. J=1,2,3,… k be the index on k-elements
4. Assign maximum value to each jth position as n-k+J
5. Start scanning the list from right to left, stop at location where there are consecutive digits such as an-1 <an. Let m = n-1. If no such pair formed, go to step 8
6. Leave the digits before am as they were, follow am by am+1, am+2…. And until k-digits in all have been listed down such that no digit crosses the limit assigned as in step 4
7. Go to step 5
8. Stop

Example 1: Find the next layer 4-combinations of the set {1,2,3,4,5,6} and after {1,2,5,6}.

Solution: The last term among the terms ai with a1 = 1, a2 = 2, a3 = 5 and a4 = 6 such that a1 = 6-4+i is a2 = 2;

To obtain the next layer 4-combination increment a2 by 1 to obtain a2 = 3. Then set a3 = 3+1 = 4 and a4 = 3+2 = 5. Hence, the next layer -4 combination is [1 3 4 5]

Discrete Probability

The word probability denotes chance and the theory of probability deals with laws governing the chances of occurrence of phenomena, which are unpredictable in nature. The theory of probability was first developed in the seventh century when certain gambling games were analyzed by the French Mathematician Blaise Pascal. Probability theory today has many applications for beyond the games of chance. Its utility is in almost all branches of Science, Engineering, Economics, Business, industry.

The theory of probability plays a vital role in making decisions in situation where there is a lack of certainty.

Some definitions in probability theory

Random Experiment: An operation which can produce some well defined outcome is known as experiment. An experiment whose outcome cannot be determined in advance in a random experiment. For example, tossing a coin is an random experiment because if a coin is tossed then either head or tail will turn up.

Example of random experiment is: Tossing a coin, throwing a die etc.

Sample Space: The set of all possible outcomes in a random experiment is called a sample space. Each element of a sample space is called a sample point.
The number of sample points in S may be denoted by n(S).
For example, in the rolling of die, the sample space S = {1,2, 3, 4, 5, 6}
Event: The results or outcomes of experiments are called events. An event is subset of sample space. For example, for the sample space in tossing two coins, a subset E = {HH, HT} is event. The number of sample points in an event E is denoted by n(E).
Equally Likely Events: A number of events are said to be equally likely if any of them cannot be expected to occur in preference to the other
Exhaustive Events: Events are said to be exhaustive when they include all possible outcomes of a random experiment. In tossing a coin, exhaustive events are two
Mutually Exclusive Events: Two or more events are said to be mutually exclusive or disjoint if the events cannot occur simultaneously. That is occurrence of one of the events prevents the occurrence of others.

Probability of Events

Every event associated with a random experiment is assigned a weight or measure of the change of it occurring called its probability. The probability of an event E is denoted by P(E) and defined as

$$P(E) = \frac{Number\ of\ outcome\ favourable\ to\ the\ occurrence\ of\ E}{Total\ number\ of\ all\ possible\ outcomes} = \frac{n(E)}{n(E)}$$

The probability of non-occurrence of the event A is

$$= \frac{Number\ of\ outcome\ un\ favourable\ to\ the\ occurrence\ of\ E}{Total\ number\ of\ all\ possible\ outcomes} = \frac{n(s)-n(E)}{n(S)} = 1-\frac{n(E)}{n(S)} = 1-P(E)$$

Example 1: Three coins are tossed

a. Write the sample space S
b. Find the probability of all heads, at most two heads

Solution: Let S denotes the sample space of tossing three coins

(a) S = {HHH, HHT, HTH, THH, HTT, TTH, THT, TTT}

Therefore n(S) = 8

(b) Let E1 be the set of favorable events that all are heads
E1 = {HHH} and n(E1) = 1
Therefore P(E1) = n(E1)/n(S) = 1/8

Let E2 be the set of favorable events that at most two heads
E2 = {HHH, HTH, THH, HTT, TTH, THT, TTT} and P(E2) = 7
Therefore P(E2) = n(E2)/n(S) = 7/8

Additional Principle

For any events A and B,
P(AUB) = P(A)+P(B) –P(A∩B)

Example: Suppose a student is selected at random 200 students, where 70 are taking C programming, 50 are taking Java and 20 taking both. Find the probability P that the student is taking C or Java.
Solution: Let C: Students taking C programming
J: Students taking Java

Therefore P(C) = $\frac{70}{200}$

P(j) = $\frac{50}{200}$

P(C∩J) = $\frac{70}{200}$

Then P = P(CUJ) = P(C)+P(J) – P(C∩J)

= $\frac{70}{200} + \frac{50}{200} - \frac{20}{200} = \frac{1}{2}$

Conditional Probability

Let E be an event in a sample space S with P(E)>0 then probability than an event A occurs once E has occurred or specifically, the conditional probability of A given E, written as P(A/E), is defined as,

P(A/E) = $\frac{P(A \cap E)}{P(E)}$

Example: What is the conditional probability that a randomly generated bit string of length four contains at least two consecutive 0s, given that the first bit is a 1? (Assume the probability of a and a 1 are the same).

Solution:

Let E be the event that a bit string of length four contains at least two consecutive 0s and F be the event that the first bit is a 1. The probability that a bit string of length four has at least two consecutive 0s, given that its first bit is a 1 equals P(E/F).

Without restriction the number of ways of bit string of length four can be formed with digits 0 and 1 is 2^4 = 16.

Keeping 1 fixed at the beginning, there are 8(23) bit strings of length four that starts with a 1. Thus P (F) = $\frac{8}{16} = \frac{1}{2}$.

Again EF = {1000, 1100, 1001}

So, P(EF) = 3/16

Consequently

P (E/F) = $\frac{P(E \cap F)}{P(F)} = \frac{3/16}{8/16} = \frac{3}{8}$

Independence

Events A and B in a probability space S are said to be independent if the occurrence of one of them does not influence the occurrence of the other. In other words, B is independent of A if P(B) is the same as P(B/A).

I.e Events are independent if P(A∩B) = P(A).P(B)

For example, in tossing of two coins, the appearance of head on one coin does not affect the appearance of head on second coin.

Example: Suppose A is the event that a randomly generated bit string of length four begins with a 1 and B is the event that this bit string contains an even number of ones. Are A and B independent, if the 16 bit strings of length four are equally likely.

Solution: There are eight bit strings of length that begin with a one: 1000, 1001, 1010, 1011, 1100, 1101, 1110 and 1111

There are also eight bit strings of length four that contains an even number of ones: 0000, 0011, 0101, 0110, 1001, 1010, 1100, and 1111.

Because A B = {1111, 1100, 1010, 1001}, we that

$P(A \cap F) = 4/16 = ¼$
Since $P(A \cap B) = ¼ = ½*1/2 = p(A).p(B)$
Hence, A and F are independent

Random variable

Many problems are concerned with a numerical value associated with the outcome of an experiment. For instance, we may want to know the probability that there are nine one bits generated when ten bits are randomly generated. To study problems of this type we introduce the concept of random variables.

A variable whose numerical value is determined by the outcome of a random experiment is called a random variable. A random variable X is a real valued function, X (x, of the elements of the sample space S where x is an element of the sample space. Random variable assigns a real number to each possible outcome. It should be noted that the range of the random variable will be a set of real numbers.

Example: If we toss a coin denote the head by 1 and tail by 0, then the random variable X takes only two values 1 and 0. Symbolically, the random variable,

$X(x) = \{x: x= (0,1) \in S\}$

Expected Value and Variance

The expected value of random variable is the sum of overall elements in a sample space of the product of the probability of the element and the value of the random variable at this moment.

The expected value of a random variable provides a central point for the distribution of the values of this random variable.

Another useful measure of a random variable is its variance, which tells how spread out the values of this random variable is:

Therefore, the expected value of the random variable X(s) on the sample space S is equal to

$E(X) = \sum_{s \in S} P(s)X(S)$

Example: Let X be the number that occurs up when a die is rolled. What is expected value of X?

Solution: The random variable X takes the values 1, 2, 3, 4, 5 or 6. What is expected value of X?

The random variable X takes the values 1, 2, 3, 4, 5 or 6 each of them has probability 1/6

This results: $E(X) = \frac{1}{6}*1+\frac{1}{6} * 2 + \frac{1}{6} * 3 + \frac{1}{6} * 4 + \frac{1}{6} * 5 + \frac{1}{6} * 6 = \frac{7}{2}$

Example: An unbiased coin is tossed three times. Let S be the sample space of the eight possible outcomes and X be the random variable that assigns to an outcome the number of tails in this outcome. What is the expected value of X?

Solution: When an unbiased coin is tossed three times, the eight possible outcomes in sample space are:

S = {HHH, HHT, HTH, THH, TTH, THT, HTT, TTT}.

Since coin is unbiased and toss is independent then the probability of each outcome is 1/8

Therefore E(x) = 1/8[X(HHH)+X(HHT)+X(HTH)+X(THH)+X(THT)+X(XTT)+X(TTT)

=1/8(0+1+1+1+2+2+2+3)

=12/8

=3/2

Example: Suppose an element X and list of n distinct real numbers are given as input then the average case complexity of linear search algorithm is the probability that X is in the list is P and it is equally likely that X is any of n element in the list.

Solution: The linear search algorithm locates and element X by successively comparing it to each element in the list and terminates when X is located and when all the elements have searched and it has been determined that X is not present in the list. 2i+1 comparisons are used if X equals the ith element of the list and 2n+2 comparisons are used if X is not in the list. Then the probability that X equals ai is p/n and probability that X is not in the list is q=1-p.

It follows that the average case complexity of linear search algorithm is:

$E = \frac{3p}{n}+\frac{5p}{n}+\ldots..+(2n+1)\frac{p}{n}+(2n+2)q$

$= \frac{p}{n}(3+5+\ldots.+(2n+1))+(2n+2)q$

$= \frac{p}{n}((n+1)^2-1)+(2n+2)q$

$= p(n+2)+(2n+2)q$

Variance

The expected value of a random variable tells us its average but nothing about how widely its values are distributed. So, the variance of a random variable helps characterize how widely a random variable is distributed.

Let X be a random variable on a sample space S. The variance of X, denoted by V(X) is

$V(X) = \sum_{s\in S}(X(S) - E(X)2P(S)$

Example: What is the variance of the random variable X, where X is the number that comes up when a die is rolled?

Solution: We have $V(X) = E(X^2)-E(X)^2$

The random variable X takes the values 1, 2, 3, 4, 5,or 6 each of them has probability 1/6.

This results: E(X): $\frac{1}{6}.1+\frac{1}{6}.2+\frac{1}{6}.3+\frac{1}{6}.4+\frac{1}{6}.5+\frac{1}{6}.6 = \frac{21}{6} = \frac{7}{2}$

Since X2 takes the values i2, i=1,2,3,4,5,6 each with probability $\frac{1}{6}$.

It follows that

$E(X2) = 1/6(1^2+2^2+3^2+4^2+5^2+6^2) = 91/6$

Hence $V(X) = \frac{91}{6}-(\frac{7}{2})^2 = \frac{35}{12}$

Randomized Algorithm

An algorithm that uses random numbers to decide what to do next anywhere in its logic is called Randomized Algorithm. For example, in Randomized Quick Sort, we use random number to pick the next pivot

Example: Algorithm to find an element 'a' in an array of n elements.

Input: An array of n≥2 elements, in which half are 'a' s and the other half are 'b' s.

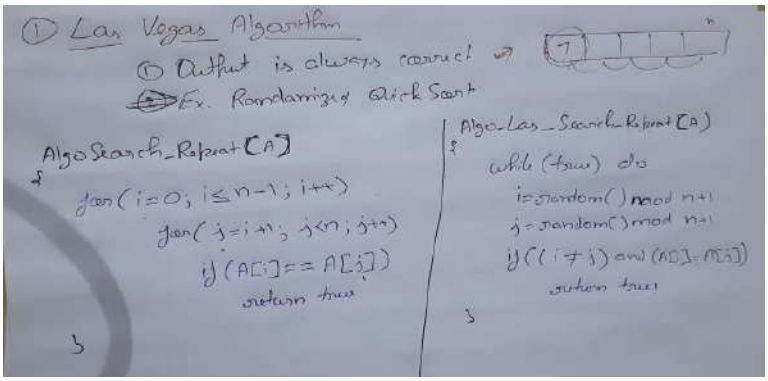
Output: To find 'a' in the array

There are two randomized algorithm to solve it

a. Las Vegas Algorithm
b. Monte Carlo Algorithm

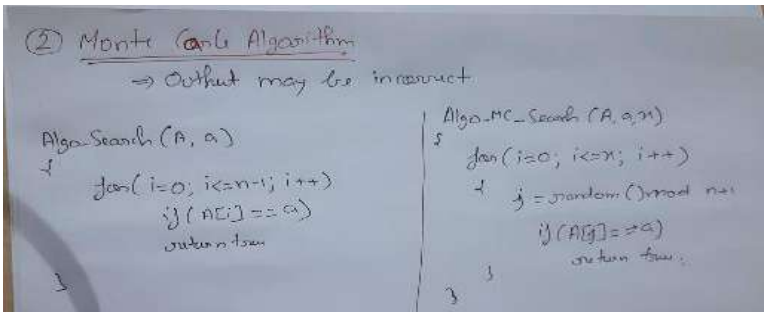
Las Vegas Algorithm: Pseudo code

finding_LV(array A, n)
begin:
    repeat
    randomly select one element out of n elements until ‘a’ is found
end



Monte Carlo Algorithm

findingA_MC (array A, n, k)
begin:
    i=0
    repeat
    randomly select one element out of n elements
    i = i+1;
    until i=k or ‘a’ is found
end



How to analyze Randomized Algorithm?

Some randomized algorithms have deterministic time complexity. For example, this implementation of Karger’s algorithm has time complexity as O(E). Such an algorithms are called Monte Carlo Algorithms and are easier to analyze for worse case

One the other hand, time complexity of other randomized algorithms (other than Las Vegas) is dependent on value of random variable. Such randomized algorithms are called Las Vegas Algorithms. These algorithms are typically analyzed for expected worst case. To compute expected time taken in worst case, all possible values of the used random variable needs to be considered in worst case and time taken by every possible value needs to be evaluated. Average of all evaluated times is the expected worst case time complexity. Below facts are generally helpful in analysis of such algorithms.

Advanced Counting

Recurrence Relation

Suppose a_0, a_1, a_2, a_3….a_n is a sequence. A recurrence relation for the nth term an is a formula giving an in terms of some of all previous terms. To find the complete sequence, the rest few initial conditions are needed. These initial values are called the initial conditions. If a recurrence relation with initial conditions is given then we can write down as many terms of the sequence as we want. If we just keep applying the recurrence.

The example, f_0=1 and f_1 = 1 are the initial condition with recurrence relation $f_n = f_{n-1}+f_{n-2}$ for n>2

The terms of sequence are 1, 1, 2, 3, 5, 8, 13 where each subsequent term is the sum of the preceding two terms. On the other hand, if you are given a sequence, you may or may not be able to determine a recurrence relation with initial conditions which describes it.

Example 1: If the given recurrence relation is $a_n = a_{n-1}+a_{n-2}$ with a_0 = 3 and a_1=5. Find the terms a_2 and a_3

Solution: Here, $a_n = a_{n-1}+a_{n-2}$

Then, $a_2 = a_1+a_0$ = 5-3=2

And $a_3 = a_2+a_1$ = 2-5 = -3

Example 2: Find the recurrence relation with initial conditions of the sequence 3,8,13, 18…

Solution: Here, a1 = 5

$a_2 = a_1+5$ = 8

$a_3 = a_2+5$=13

$a_4 = a_3+5$ = 18

……

$a_n = a_{n-1}+5$

Hence, the recurrence relation is

$a_n = a_{n-1}+5$, n_2, a1=3

Example 3: Find the recurrence relation of the sequence 2, 5, 11, 23, 47

Solution:

Here a_1 = 2

$a_2 = 5 = 2*a_1+1$
$a_3 = 11 = 2*a_2+1$
$a_3 = 23 = 2*a_3+1$
……..
$a_4 = 2a_{n-1}+1$
Hence, the recurrence relation is $a_4 = 2a_{n-1}+1$ with initial conditions $a_1 = 2$, $n\geq2$
Example 4: Find the recurrence of the sequence 1, 1, 2, 3, 5, 8, 13 ….
Solution:
Here $a_1 = 1$
$a_2 = 1$
$a_3 = 2 = a_1+a_2$
$a_4 = 3 = a_3+a_2$
$a_5 = a_4+a_3$
…….
$a_n = a_{n-1}+a_{n-2}$
Hence, the recurrence relation is
$a_n = a_{n-1}+a_{n-2}$ with initial conditions $a_1 = 1$, $a_2 = 1$ for $n\geq3$

Solving recurrence Relations

If the given recurrence relation involving sequence a_0, a_1, a_2….a_n then the solution of such recurrence relation is to find an explicit formula for the general term a_n
To solve the recurrence relation, we may replace each of a_{n-1}, a_{n-2},…… by their predecessors. This process continues until an explicit formula for nth term a_n.

Example 5: Solve the recurrence relation $a_n = 2n-1$, $n\geq1$ and a0=3
Solution:
$a_0 = 3$
$a_1 = 2a_0 = 2.3$
$a_2 = 2a_1 = 2(2.3) = 2^2.(3)$
$a_3 = 2a_2 = 2(2^2.3) = 2^3(3)$
$a_4 = 2a_3 = 2(2^3.3) = 2^4(3)$
…….
$a_n = 2^n(3)$
Hence, general solution for the given recurrence relation is $an = 2^n(3)$
Example 6: Solve the recurrence relation $a_n = a_{n-1}+2$ subject to initial condition $a_1 = 3$
Solution:
$a_n = a_{n-1}+2$ step 0
$=(a_{n-2}+2)+2$ step 1
$=(a_{n-3}+2)+4$ Step 2
$=(a_{n-4}+2)+6$ step 3
$=(a_{n-5}+2)+8$ step 4

=……

$a_n = a_1+2(n-1)$ step n-1

Example 7: Show that solution of Recurrence relation $a_n = 2a_{n-1}- a_{n-2}$ is $a_n = 3n$

Solution:

$a_n = 2a_{n-1}-a_{n-2}$

$a_n = 2[3(n-1)-3(n-2)]$

$a_n = 2(3n-3)-3n+6$

$a_n = 6n-6-3n+6$

$a_n = 3n$

Solving linear recurrence relations

A wide variety of recurrence relations occur in models. Some of these recurrence relations can be solved using iteration or some other ad hoc technique. However, one important class of recurrence relations can be explicitly solved in a systematic way. These are recurrence relations that express the terms of a sequence as linear combination of previous terms.

Definition:

A linear homogenous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$a_n = c_1a_{n-1}+c_2a_{n-2}\ldots..c_ka_{n-k}$

where $c_1,c_2,c_3\ldots.\ c_k$ are real numbers and $c_k\neq 0$

The recurrence relation in the definition is linear because the right hand side is sum of previous terms of the sequence each multiplied by a function of n. The recurrence relation is homogeneous because no terms occur that are not multiples of ajs. The coefficients of the terms of the sequence are all constants; rather than functions that depend on n. The degree is k because a_n is expressed in terms of previous k terms of the sequence

Example 1: The recurrence relation $P_n = \{1.11\}P_{n-1}$ is a linear homogeneous recurrence relation of degree one. The recurrence relation $f_n = f_{n-1}+f_{n-2}$ is a linear homogeneous recurrence relation of degree two.

The recurrence relation $a_n = a_{n-1}+a^2_{n-2}$ is not linear.

General form of Linear Homogeneous Recurrence Relation

The homogeneous recurrence of degree k_1 with constant coefficient has the general form

$a_n = c_1a_{n-1}+c_2a_{n-2}+\ldots+c_{k.}a_{n-k}$……..(i)

If $a_n = r^n$ is a solution of equation (i) then it must satisfy equation (i) i.e.

$r^n = c_1r^{n-1}+c_2r^{n-2}+\ldots+c_kr^{n-k}$

Dividing both sides r^{n-k} on both side we get

$r^k = c_1r^{k-1}+c_2r^{k-2}+\ldots..c_k$

or $r^k-c_1r^{k-1}c_2r^{k-2}\ldots\ldots-c_k=0$

This equation is known as characteristic equation of given recurrence relation and it provides characteristics roots of recurrence relation which are used to give on explicit formula for all the solution of recurrence relation

Solving Linear Homogeneous Recurrence Relations with Constant Coefficients

A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form $a_n = c_1a_{n-1}+c_2a_{n-2}+\ldots\ldots\ldots\ldots+c_ka_{n-k}$, where c_1, c_2, c_2, …. are real numbers and $c_k \neq 0$. The above relation is linear since right had side is sum of the multiples of previous terms of the sequence. It is homogeneous since no term occurs without being multiple of some aj,s. All the coefficients of the terms are constants and degree k is due to the representation of in terms of previous 'k' terms of sequence.

In solving recurrence relation of this type, the approach is to look for the solution of the form an = rn where r is a constant. $a_n = r^n$ is a solution of a recurrence relation $a_n = c_1a_{n-1}+c_2a_{n-2}+\ldots+c_k.a_{n-k}$ if and only if $r^n = c_1r^{n-1}+c_2r^{n-2}+\ldots+c_kr^{n-k}$

When we divide both sides by r^{n-k} and transpose the right hand side we have

$r^k-c_1r^{k-1}c_2r^{k-2}\ldots\ldots-c_k=0$

Here we say $a_n = r^n$ is a solution if and only if r is the solution if the equation $r^n-c_1r^{k-1}-c_2r^{k-2}-\ldots-c_k = 0$ (characteristic equation of the recurrence relation) and solutions to this equations are called characteristic root of the recurrence relation.

Theorem 1 (Without Proof): Let c_1 and c_2 be real numbers. Suppose that $r^2-c_1r-c_2 =0$ has two distinct roots r_1 and r_2. Then the sequence {an} is a solution of the recurrence relation $a_n = c_1a_{n-1}c_2a_{n-2}$ if and only if $a_n = \alpha_1r_1^n+ \alpha_2r_2^n$ for n=0, 1, 2, 3…. where α_1 and α_2 are constants.

Example 1: Solve the recurrence relation $a_n = 5a_{n-1}-6a_{n-2}$ for $n\geq 2$, $a_0 = 1$ and $a_1 = 0$

Solution: The given recurrence relation is: $a_n = 5a_{n-1}-6a_{n-2}$ …..(i)

The characteristics equation is

$r^2-5r+6=0$

$r^2 – 3r-2r+6=0$

$r(r-3)-2(r-3)=0$

$(r-2)(r-3) = 0$

$r = 2,3$

$r_1 = 2$, $r_2 = 3$

Since, two characteristics roots are distinct; we use the "theorem 1" to write the general solution

The general form of the solution is:

$a_n = \alpha_1r_1+ \alpha_2r_2^n$

or $a_n = \alpha_12^n+ \alpha_23^n$ ……(ii)

Form initial conditions

$a_0 = \alpha_1 2^0 + \alpha_2 3^0$
$1 = \alpha_1 + \alpha_2$
And
$a_1 = \alpha_1 2^1 + \alpha_2 3^1$
$0 = 2\alpha_1 + 3\alpha_2$

Solving we get,
$\alpha_1 = 3$ and $\alpha_2 = -2$

Therefore, the solution of given recurrence relation is

$$a_n = 3.2^n + (-2)3^n$$

Do now :Example 2: Solve the recurrence relation $a_n = 6a_{n-1} - 8a_{n-2}$ for $n \geq 2$, $a_0 = 4$ and $a_1 = 10$

Solution:

The given recurrence relation is
$a_n = 6a_{n-1} - 8a_{n-2}$
The characteristic equation is
$r^2 - 6r + 8 = 0$
$r^2 - 4r - 2r + 8 = 0$
$r(r-4) - 2(r-4) = 0$
$(r-4)(r-2) = 0$
i.e. $r = 4, 2$
$\rightarrow r1 = 4$ and $r2 = 2$
Since roots are distinct, the general form of solution is:
$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$
Or $a_n = \alpha_1 4^n + \alpha_2 2^n$
From initial conditions, we have
$a_0 = \alpha_1 4^0 + \alpha_2 2^0$
or $4 = \alpha_1 + \alpha_2$……..(i)
And
$a_1 = \alpha_1 4^1 + \alpha_2 2^1$
$10 = 4\ \alpha_{1+2}\ \alpha_2$…….(ii)

Solving equation (i) and (ii), we get
$\alpha_1 = 1$ and $\alpha_2 = 3$
Therefore, the solution of given recurrence relation is
$a_n = 1.4^n + 3.2^n$

Theorem 2: (Without proof): Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0. Then the sequence {an} is a solution of the recurrence relation $an = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $an = \alpha_1 r_0^n + \alpha_2 r_0^n$ for n=0, 1, 2,….. where α_1 and α_2 are constants.

Example 1: Solve the recurrence relation an = $6a_{n-1} - 9a_{n-2}$ for $n \geq 2$, $n_0 = 1$, $a_1 = 6$

Solution: Characteristic equation of the given relation is:

$r^2-6r+9=0$

$(r-3)^2 = 0$

$r = 3, 3$

Therefore its only roots are = 3, 3

Hence, the sequence {an} is a solution to the recurrence relation if and only if

$a_n = \alpha_1 3^n+\alpha_2{}^3$ for some constants α_1 and α_2

From initial conditions we have

$a_0 = 1 = \alpha_1+\alpha_2$

$a_1 = 6 = 3\alpha_1+3\alpha_1$

Solving these two equations we have $\alpha_1 = 1$ and $\alpha_2 = 1$

Hence, the solution is the sequence $\{a_n\}$ with

$a_n = 1.3^n+1.n.3^n$

Example 2: Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, $a_0 = 4$, $a_1 = 1$

Solution:

The given recurrence relation is

$a_n = 2a_{n-1} - a_{n-2}$

The characteristic equation for this recurrence relation is

$r^2-2r+1=0$

$(r-1)^2 = 0$

$r=1,1$

Since the characteristic roots are same, the general form of solution is

$a_n = \alpha_1 r_0{}^n+\alpha_2 n.r_0{}^n$

That is

$a_n = \alpha_1 1^n+\alpha_2 n.1^n$…….(i)

From initial conditions, we have

$a_0 = \alpha_1 1^0+\alpha_2 n.1^0$

Or $4 = \alpha_1 +\alpha_2 0$

Or $\alpha_1 = 4$

And

$a_1 = \alpha_1 1^1+\alpha_2.1.1^1$

or $1 = \alpha_1+ \alpha_2$……(ii)

Putting value of $\alpha_1 = 4$ in equation (iii), we have

$1 = 4+ \alpha_2$

Or

$\alpha_2 = -3$

Therefore, the solution of given recurrence relation is

$a_n = 1. 1^n+(-3)n.1^n$

Theorem 3: (without proof)

Let c_1, c_2 c_3,…. c_k be real numbers. Suppose that $r^k-c_1r^{k-1}-\ldots-c_k=0$ has k distinct roots $r_1,r_2,r_3\ldots\ldots r_k$. Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1a_{n-1}+c_2a_{n-2}+\ldots+c_ka_{n-k}$ if and only if $a_n = \alpha_1r_1^n+\alpha_2r_2^n+\ldots\ldots\ldots\ldots\ldots+\alpha_nr_k^n$ for n=0,1,2,3….. where α_1, α_2 ….. α_k are constants.

Example 1: Find the solution to $a_n = 2a_{n-1}+a_{n-2}-2a_{n-3}$ for $n\geq3$ with $a_0 = 3$, $a_1 = 6$ and $a_2 = 0$

Solution: The given recurrence relation is:

$a_n = 2a_{n-1}+a_{n-2}-2a_{n-3}$ ……..(i)

The characteristics equation is:

$r^3 - 2r^2 - r + 2 = 0$

$r^3 - r^2 - r^2 + r - 2r + 2 = 0$

$r^2(r-1) - r(r-1) - 2(r-1) = 0$

$(r-1)(r^2 - r-2) = 0$

$(r-1)(r2 - 2r + r - 2) = 0$

$(r-1)(r(r-2)+1(r-2)) = 0$

$(r-1)(r-2)(r+1) = 0$

Therefore, $r_1 = 1$, $r_2 = 2$, $r_3 = -1$

Since all three roots are different, we can write the general form of solution as:

$a_n = \alpha_1r_1^n + \alpha_2r_2^n + \alpha_3r_3^n$

or $a_n = \alpha_1 1^n + \alpha_2 2^n + \alpha_3(-1)^n$ ……(ii)

From given initial conditions, we have

$a_n = \alpha_1r_1^n + \alpha_2r_2^n + \alpha_3r_3^n$

$3 = \alpha_1 1^0 + \alpha_2 2^0 + \alpha_3(-1)^0$

Again

$a_1 = \alpha_1.1 + \alpha_2.2 + \alpha_3(-1)$

or $6 = \alpha_1 + 2\alpha_2 - \alpha_3$

and

$a_2 = \alpha_1.1^2 + \alpha_2.2^2 + \alpha_3(-1)^2$

or $0 = \alpha_1 + 4\alpha_2 + \alpha_3$

Solving we get

$\alpha_1 = 6$

$\alpha_2 = -1$

$\alpha_3 = -2$

Therefore the solution of given recurrence relation is:

$a_n = \alpha_1r_1^n + \alpha_2r_2^n + \alpha_3r_3^n$

$a_n = (-1).1^n + 6 . 2^n - (-2)(-1)^n$

Example 2: Solve the recurrence relation $a_n = 2a_{n-1}+a_{n-2}-2a_{n-3}$ for $n\geq3$, $a_0=3$ $a_1 = 6$ and $a_2=9$

Solution:

The given recurrence relation is:

$a_n = 2a_{n-1}+a_{n-2}-2a_{n-3}$

The characteristic equation of this recurrence relation is

$r^3-2r^2-r+2=0$

$r^3-r^2-r^2+r-2r+2=0$

$r^2(r-1)-r(r-1)-2(r-1) = 0$
$(r-1)(r^2-r-2)=0$
$(r-1)(r^2-2r+r-2)=0$
$(r-1)(r(r-2)+1(r-2))=0$
$(r-1)(r+1)(r-2)=0$
The roots of this characteristic equation are
$r=1, -1, 2$
Hence the sequence $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_1 1^n + \alpha_2(-1)^n + \alpha_3 2^n$ for some constants α_1, α_2 and α_3
From initial conditions we have,
$a_0 = 3 = \alpha_1 + \alpha_2 + \alpha_3$
$a_1 = 6 = \alpha_1 - \alpha_2 + 2\alpha_3$
And
$a_2 = 9 = \alpha_1 + \alpha_2 + 4\alpha_3$
Solving these equations we have,
$\alpha_1 = 3/2$
$\alpha_2 = -1/2$
$\alpha_3 = 2$
Hence, the solution is the sequence $\{a_n\}$ with
$a_n = (3/2)1^n + (1/2)(-1)^n + 2.2^n$

Theorem 4 (Without proof)
Let $c_1, c_2, c_3, c_4 \ldots\ldots c_k$ be real numbers. Suppose that $r^k - c_1 r^{k-1} - \ldots . c_k = 0$ has distinct roots $r_1, r_2, r_3 \ldots r_k$ with multiplicity $m_1.m_2, m_3 \ldots m_k$ respectively such that $m_i \geq 1$ for $i=0,1,2,3\ldots t$ and $m_1+m_2+m_3+\ldots+m_t = k$
Then the sequence {an} is a general solution of the recurrence relation
$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots\ldots\ldots + c_k a_{n-k}$ if and only if
$a_n = (\alpha_{1,0} + \alpha_{1,1} n + \ldots\ldots + \alpha_{1,m1-1} n^{m1-1}) r_1{}^n + (\alpha_{2,0} + \alpha_{2,1} n + \ldots\ldots + \alpha_{2,m2-1}.n^{m2-1}) r_2{}^n + \ldots\ldots\ldots\ldots\ldots\ldots$
$\ldots + ((\alpha_{t,0} + \alpha_{t,1} n + \ldots\ldots + \alpha_{t,mt-1} n^{mt-1}) rt^n$
For $n=0, 1, 2\ldots\ldots\ldots\ldots$ where $\alpha_{i,j}$ are constants for $1 \leq j \leq m_i - 1$
Example 1: Solve the recurrence relation $a_n = 5a_{n-1} - 7a_{n-2} + 3a_{n-3}$ for $n \geq 3$, $a_0 = 1$, $a_1 = 9$ and $a_2 = 15$
Solution: Characteristic equation of the given relation is $r^3 - 5r^2 + 7r - 3 = 0$.
Its roots are r1 = 1, r2=3 and r = 1
That is, r1 = 1 with m1 = 2 and r2 = 3 with m2=1
Hence, the sequence {an} is a solution to the recurrence relation if and only if $an = (\alpha_{1,0} + \alpha_{1,1} n)1n + (\alpha_{2,0})3n$, for some constants $\alpha_{1,0}$, $\alpha_{1,0}$, $\alpha_{2,0}$
From the initial conditions we have
$a0 = 5 = \alpha_{1,0} + \alpha_{2,0}$
$a1 = 9 = \alpha_{1,0} + 3\,\alpha_{2,0}$
$a2 = 15 = \alpha_{1,0} + 2\,\alpha_{1,1} + 9\,\alpha_{2,0}$
Solving these three equations, we get,
$\alpha_{1,0} = 3/2$
$\alpha_{1,1} = 9$
$\alpha_{2,0} = -1/2$
Hence, the solution is the sequence {an} with

$a_n = (3/2)1^n + 91^n - (1/2)3^n$

Example 2: Solve the recurrence relation $a_n = -3a_{n-1}-3a_{n-2}-a_{n-3}$ with $a_0=-5$, $a_1 = 4$ and $a_2=15$

Solution: The given recurrence relation is:

$a_n = -3a_{n-1}-3a_{n-2}-a_{n-3}$

The characteristic equation is:

$r^3+3r^2+3r+1=0$

$(r+1)^3=0$

$r = -1,-1,-1$

Since all these roots are same with multiplicity m=3

The general solution is:

$a_n = (\alpha_1+ \alpha_1 n+ \alpha_3 n^2)r^n$

$a_n = (\alpha_1+ \alpha_1 n+ \alpha_3 n^2)(-1)^n$

From given initial condition

$a_0 = (\alpha_1+ \alpha_1 0+ \alpha_3 0^2)(-1)^0$

or $-5 = (\alpha_1+ 0+ 0)1$

Or $\alpha_1 = -5$……..(i)

Again,

$a_1 = (\alpha_1+ \alpha_1 1+ \alpha_3 1^2)(-1)^1$

or

$4 = (\alpha_1+ \alpha_1 1+ \alpha_3 1^2)(-1)^1$

or $\alpha_1+ \alpha_1+ \alpha_1 = -4$……..(ii)

And

$a_2 = (\alpha_1+ \alpha_1 2+ \alpha_3 2^2)(-1)^2$

or

$a_2 = (\alpha_1+ \alpha_1 2+ \alpha_3 2^2)(-1)^2$

$15 =(\alpha_1+2 \alpha_2+4 \alpha_3)(-1)^2$

Or $\alpha_1+2 \alpha_2+4 \alpha_3 = 15$…….(iii)

Solving equation (i), (ii) and (iii), we get

$\alpha_1 = -5$

$\alpha_2 = -8$

$\alpha_3 = 9$

Therefore, the solution of given recurrence relation is:

$a_n = (-5- 8n+ 9n^2)(-1)^n$

Solving Linear Non-homogeneous Recurrence Relations with Constant Coefficients

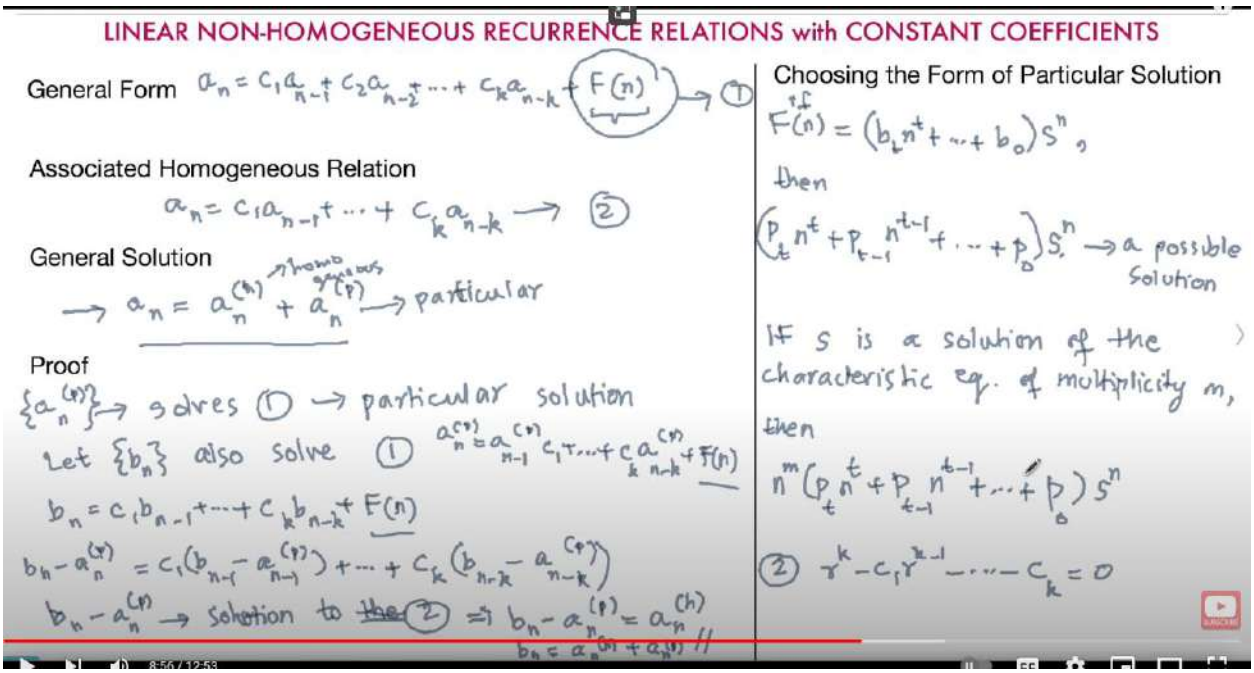
The recurrence relation of the form

$a_n = c_1a_{n-1}+c_2a_{n-2}+\ldots..+c_ka_{n-k}+ F(n)$

where c_1, c_2, c_3,….. c_k are real numbers and F(n) is a function depending upon n. The recurrence relation preceding F(n) is called associated homogeneous recurrence relation

For example,

$a_n = 7a_{n-1}+3a_{n-2}+6n$ is a linear non-homogeneous recurrence relation with constant coefficients.



Theorem 5: (without proof)

If $\{a_n^{(p)}\}$ is a particular solution of the non-homogeneous linear recurrence relation with constant coefficients $a_n = c_1a_{n-1} + c_2a_{n-2}+\ldots+c_ka_{n-k} + F(n)$, then every solution of the form $\{a_n^{(p)}+a_n^{(h)}\}$,

where $a_n^{(h)}$ is a solution of the associated homogeneous recurrence relation $a_n = c_1a_{n-1} + c_2 a_{n-2} +\ldots\ldots+ c_ka_{n-k}$

Example 1: Find all solutions of the recurrence relation $a_n = 4a_{n-1}+ n^2$. Also find the solution of the relation with initial condition $a_1 = 1$

Solution:

We have associated linear homogeneous recurrence relation as $a_n = 4a_{n-1}$. The root is 4, so the solutions are $a_n(h) = \alpha 4^n$ where α is a constant. Since $F(n) = n^2$ is a polynomial of degree 2, a trial solution is a quadratic function in n, say, $p_n = an^2+bn+c$, where a,b,c are constants.

To determine whether there are any solutions of this form, suppose that

$p_n = an^2+bn+c$ is such solution.

Then the equation $a_n = 4a_{n-1}+n^2$ becomes

$an^2+bn+c = 4(a(n-1)^2+b(n-1)+c)+n^2$

$= 4an^2 -8an +4a +4bn-4b+4c+n^2$

$= (4a+1)n^2+(-8a+4b)n+(4a-4b+4c)$

Here an^2+bn+c is the solution if and only if

$4a+1=a$. That is $a=-1/3$

$-8a+4b = b$ or $b = -8/9$

And

$4a-4b+4c =c$ or $c = -28/27$

So $a_n(p) = -1/3(n^2+8/9.n+28/9)$ is a particular solution and all other solutions are

$a_n = [a_n(p)+a_n(h)\} = 1/3(n^2+8/3n+28/9)+ \alpha 4^n$ where α is a constant

For solution with $a_1 = 1$, we have

$a_1 = 1 = -1/3(1+8/3+28/9)+\alpha.4$

Or $\alpha = 22/27$

Then the solution is

$a_n = [a_n(p)+a_n(h)\} = 1/3(n^2+8/3n+28/9)+ 22/27.4^n$

Theorem 6: (without proof)

Suppose that $\{a_n\}$ satisfies the linear non-homogeneous recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2} +\ldots + c_ka_{n-k} +F(n)$, where $c_1, c_2, c_3\ldots c_k$ are real numbers and $F(n) = (b_tn^t + b_{t-1}n^{t-1} + \ldots +b_1n +p_0)\, s^n$, where $b_0, b_1, b_2\ldots b_t$ and s are real numbers. When s is not root of the characteristic equation of the associated homogeneous recurrence relation, there is a particular solution of the form

$(p_tn^t + p_{t-1}n^{t-1}+\ldots+p_1n+p_0)s^n$

When s is the root of the characteristic equation and its multiplicity is m, there is a particular solution of the form

$n^m (p_tn^t + p_{t-1}n^{t-1}+\ldots+p_1n+p_0)s^n$

Example: Find the solution of the recurrence relation $a_n = 2a_{n-1}+n.2^n$

Solution:

We have associated linear homogeneous recurrence relation is

$a_n^{(h)} = 2a_{n-1}$

The characteristic equation for this would be r-2 =0, so the root is 2 and hence the solution is $a_n^{(h)} = \alpha_1 2^n$, where α is a constant

We have $F(n) = n2^n$ where s is the root of the characteristic equation and the multiplicity of 2 is 1, so the particular solution has the form

$c = n.(p_1n)2^n$

or $a_n(p) = p_1n^22^n$

The all solution is $a_n = \alpha_1 2^n+p_1n^22^n$

Example 2 : What form does a particular solution of the linear nonhomogeneous recurrence an = 6an-1-9an-2 +F(n) have when

a) $F(n) = 3^n$
b) $F(n) = n3^n$
c) $F(n) = n^22^n$
d) $F(n) = (n^2+1)3^n$?

Solution: The associated linear homogeneous recurrence relation is $a_n = 6a_{n-1} - 9a_{n-2}$. Its characteristic equation is, $r^2-6r+9 = (r-3)^2=0$ has a single root which is 3 with multiplicity 2. Theorem 6 with F(n) of the form $P(n)s^n$, where P(n) is a polynomial and s is a constant, we need to ask whether s is a root of this characteristic equation

a. Because s=3 is a root with multiplicity m=2. Theorem tells us that a particular solution has the form $(p_0n^23^n)$ if $F(n) = 3^n$
b. Particular solution has the form $n^2(p_1n+p_0)3^n$
c. Particular solution has the form $(p_2n^2+p_1n+p_0)2^n$
d. Particular solution has the form $n2(p_2n^2+p_1n+p_0)3^n$

Recurrence Applications

One of the application areas of recurrence relations is analysis of divide and conquers algorithms

Divide and Conquer Algorithms

Divide and conquer algorithms divide a problem of larger size to the problem of smaller size so that the problem of smallest size that has trivial solution is obtained. If f(n) represents the number of operations required to solve the problem of size n, then follows the recurrence relation f(n) = af(n/b)+ g(n), called divide and conquer recurrence relation. In the relation above, the problem of size n is partitioned into a part.

Example 1: Fibonacci Numbers

We know that the Fibonacci numbers are generated by the formula $f_n = f_{n-1} + fn_{-2}$. Here nth Fibonacci number is the sum of $(n-1)^{th}$ and $(n-2)^{th}$ Fibonacci numbers. Here, for the initial conditions are $f_0 = 0$ and $f_1 = 1$. Use of the above relation does not exactly produce the recurrence relation mentioned above. However, this is an example of divide and conquer algorithm since each time the problem is changed into two problems of smaller size

Example 2: Merge Sort

In merge sorting, the input sequence of n items is broken down into two halves. Since the list of size n need more comparisons than list of size n/2, the problem here is simplified. This process continues until all the comparisons are trivial.

This problem satisfies the divide and conquers recurrence relation

M (n) = 2M (n/2) +O (1)

Example 3: Binary Search: The binary search algorithm reduces the search for an element in a search sequence of size n to the binary search for this element in a search sequence by n/2 when n is even. The problem size n has been reduced to one problem of size n/2. So two comparisons are needed to implement this reduction. Hence if f(n) is the number of comparisons required to search for an element in a search sequence of size n, then

f(n) = f(n/2)+2